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# Practical 6: Logistic Regression

# 1. Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    ConfusionMatrixDisplay
)

# 2. Load the dataset
data = load_breast_cancer()

X = data.data
y = data.target

# 3. Display dataset
df = pd.DataFrame(X, columns=data.feature_names)
df["Target"] = y

print("First five rows:")
print(df.head())
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# 4. Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 5. Scale the features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 6. Create and train Logistic Regression model
model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)

# 7. Make predictions
y_pred = model.predict(X_test)

# 8. Calculate evaluation metrics
print("\nLogistic Regression Results")

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))

# 9. Display confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
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print(cm)


# 10. Plot confusion matrix
display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=data.target_names
)

display.plot(cmap="Blues")
plt.title("Logistic Regression Confusion Matrix")
plt.show()
```